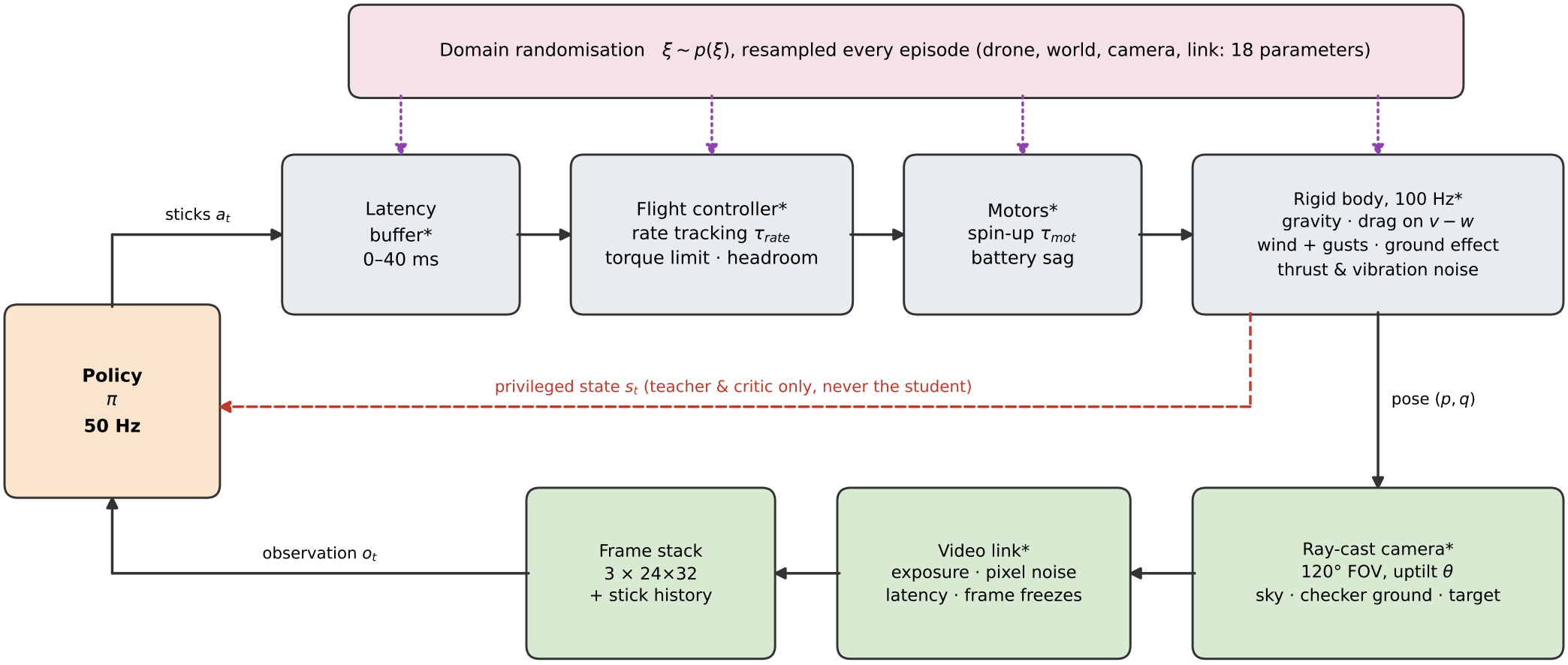
Domain randomisation $\xi \sim p(\xi)$, resampled every episode (drone, world, camera, link: 18 parameters)
sticks a_t
Latency
buffer*
0–40 ms
Flight controller*
rate tracking τ_{rate}
torque limit · headroom
Motors*
spin-up τ_{mot}
battery sag
Rigid body, 100 Hz*
gravity · drag on $v - w$
wind + gusts · ground effect
thrust & vibration noise
Policy
π
50 Hz
privileged state s_t (teacher & critic only, never the student)
pose (p, q)
observation o_t
Frame stack
3 × 24×32
+ stick history
Video link*
exposure · pixel noise
latency · frame freezes
Ray-cast camera*
120° FOV, uptilt θ
sky · checker ground · target
* depends on randomised parameters